| | | |
|---|---|---|
| | | Points between two adjacent nodes are in phase. Points in adjacent segments are anti-phase. |
| 17 | **D** | Using $x = \frac{\lambda D}{a}$,<br><br>$x = \frac{(600 \times 10^{-9})(1)}{a}$<br>$2x = \frac{(400 \times 10^{-9})D}{a}$<br>D = 3.00 m |
| 18 | **B** | Current in circuit, $I = \frac{V}{R} = \frac{10}{4.0} = 2.5\text{ A}$,<br><br>Potential difference across connecting wires<br>= 12 – 10.0 – (2.5)(0.20)<br>= 1.5 V<br><br>Power loss in connecting wires<br>= (2.5)(1.5)<br>= 3.75 W |

| | | |
|---|---|---|
| 19 | A | [illegible] |